

Averages and Spread

GCSE Maths · Practice Questions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1

[1 mark]

Write down the mode of these numbers.

5 8 5 2 9

.....

2

[2 marks]

Work out the mean of

3 9 4 8 6

.....

3

[2 marks]

Here are six numbers.

14 7 11 3 9 20

(a) Work out the median.

(b) Work out the range.

.....

4

[2 marks]

A shoe shop wants to know which size to stock the most of.

Which average should the manager use, and why?

.....

5

[3 marks]

The table shows the number of goals scored in 20 matches.

Goals	0	1	2	3
Frequency	5	8	4	3

Work out the mean number of goals per match.

6

[3 marks]

The mean of four numbers is 12.

Three of the numbers are 8, 15 and 11.

Work out the fourth number.

7

[3 marks]

Here is a set of data.

5 6 7 8 44

(a) Work out the mean.

(b) Work out the median.

(c) Which average better represents this data? Explain.

8

[3 marks]

Here are eight numbers in order.

3 5 7 8 11 13 16 20

(a) Work out the lower and upper quartiles.

(b) Work out the interquartile range.

9

[4 marks]

The table shows the times, t minutes, taken by 40 people to complete a task.

Time (t min)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 40$
Frequency	6	14	12	8

Work out an estimate for the mean time.

.....

10

[4 marks]

- Team A scored these goals in 8 matches: 0, 1, 1, 2, 3, 4, 4, 5
- Team B scored these goals in 8 matches: 1, 2, 2, 2, 3, 3, 3, 4
- (a) Work out the mean for each team.
- (b) Work out the range for each team.
- (c) Compare the two teams.

.....

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

The mean of 10 numbers is 18. Another number, 29, is added to the set.

Work out the new mean.

.....

2

[2 marks]

The mean of five numbers is 14. Four of them are 10, 12, 15 and 18.

Work out the fifth number.

.....

3

[3 marks]

The mean mark of 8 boys is 65. The mean mark of 12 girls is 70.

Work out the mean mark of all 20 students.

.....

4

[3 marks]

The table shows the times, t minutes, of 40 phone calls.

Time (t min)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 40$
Frequency	7	15	13	5

Estimate the median time.

.....

5

[4 marks]

Here are nine values in order.

12 15 18 21 22 25 28 30 35

Work out the five values needed to draw a box plot, and the interquartile range.

.....

6

[4 marks]

Box plots were drawn for two groups.

	Min	LQ	Median	UQ	Max
Group X	5	12	18	26	40
Group Y	8	14	17	21	30

- (a) Work out the interquartile range for each group.
- (b) Make two comparisons between the groups.

.....

7

[4 marks]

The table shows the number of pets owned by some families.

Pets	0	1	2	3	4
Frequency	3	5	x	4	2

The mean number of pets is 1.8. Work out the value of x .

.....

8

[4 marks]

The table shows data grouped into classes of unequal width.

Value (x)	$0 < x \leq 5$	$5 < x \leq 15$	$15 < x \leq 35$
Frequency	8	22	20

- (a) Work out an estimate for the mean.
- (b) Work out which class is the modal class, explaining your reasoning.

9

[4 marks]

A set of 12 numbers has a mean of 25.

Two of the numbers, 18 and 22, are removed.

Work out the mean of the remaining numbers.

10

[5 marks]

The table shows the number of siblings of 30 students.

Siblings	0	1	2	3	4
Frequency	6	10	8	4	2

- (a) Write down the mode.
- (b) Work out the median.
- (c) Work out the mean, to 2 decimal places.
- (d) Write down the range.